



Configuring STP to Prevent Switching Loops

Difficulty (out of 3 stars): ★★★

Recommended study time: 2 hours

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1 About This Lab

1.1 Task Description

On a switched Ethernet network, redundant links are typically used to implement link backup and enhance network availability. The use of redundant links, however, may produce loops, causing broadcast storms and MAC address table instability. As a result, the communication quality deteriorates, and the communication service may even be interrupted. Spanning Tree Protocol (STP) solves this problem.

After completing this task, you will be able to configure Huawei switches, understand STP, and know how to use STP to resolve loop issues on a switched network.

1.2 Objectives

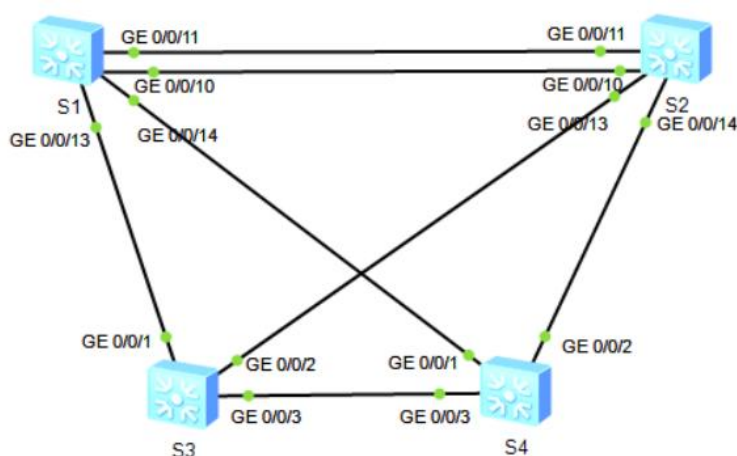
On completing this task, you will be able to:

1. Learn how to enable and disable STP/RSTP.
2. Learn how to change the STP mode of a switch.
3. Learn how to change bridge priorities to control root bridge election.
4. Learn how to change port priorities to control the election of the root port and designated port.
5. Learn how to change port costs to control the election of the root port and designated port.
6. Learn how to configure edge ports.
7. Learn how to enable and disable RSTP.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology



A company uses redundant links on its Layer 2 switched network to improve network reliability. To prevent broadcast storms and MAC address flapping caused by redundant links, deploy STP on switches.



2.2 Initial Configuration

- Initial configuration of S1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S1
[S1]
```

- Initial configuration of S2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S2
[S2]
```

- Initial configuration of S3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S3
[S3]
```

- Initial configuration of S4

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname S4
[S4]
```

3 Task Implementation

3.1 Checking Basic Configurations

The details are not provided.

3.2 Enabling STP

1. Enable STP globally.

```
[S1]stp enable
```

The **stp enable** command enables STP, RSTP, or MSTP on a switching device or a port. By default, STP, RSTP, or MSTP is enabled on switches.

2. Change the spanning tree mode to STP.

```
[S1]stp mode stp
```

Info: This operation may take a few seconds. Please wait for a moment...done.

The **stp mode { mstp | rstp | stp }** command sets the spanning tree working mode on a switching device. By default, devices operate in MSTP mode. The spanning tree mode of the current device has been changed to STP.

```
[S2]stp enable
```

```
[S2]stp mode stp
```

Info: This operation may take a few seconds. Please wait for a moment...done.



```
[S3] stp enable
[S3]stp mode stp
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S4] stp enable
[S4]stp mode stp
Info: This operation may take a few seconds. Please wait for a moment...done.
```

3. Check the spanning tree status. S1 is used as an example.

```
[S1]display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge          :32768.4c1f-cc33-7359           //Bridge ID of the
device
Config Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC       :32768.4c1f-cc10-5913 / 20000    //ID of the current root
bridge and root path cost
CIST RegRoot/IRPC    :32768.4c1f-cc33-7359 / 0
CIST RootPortId      :128.14
BPDU-Protection      :Disabled
TC or TCN received   :47
TC count per hello   :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:0m:38s
Number of TC          :15
Last TC occurred      :GigabitEthernet0/0/14
```

The command also displays port status information, which is not included in the preceding output.

4. Check the brief spanning tree information on each switch.

```
[S1]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/10  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/11  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/13  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/14  ROOT    FORWARDING   NONE
```

```
[S2]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/10  ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/11  ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/13  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/14  ROOT    FORWARDING   NONE
```

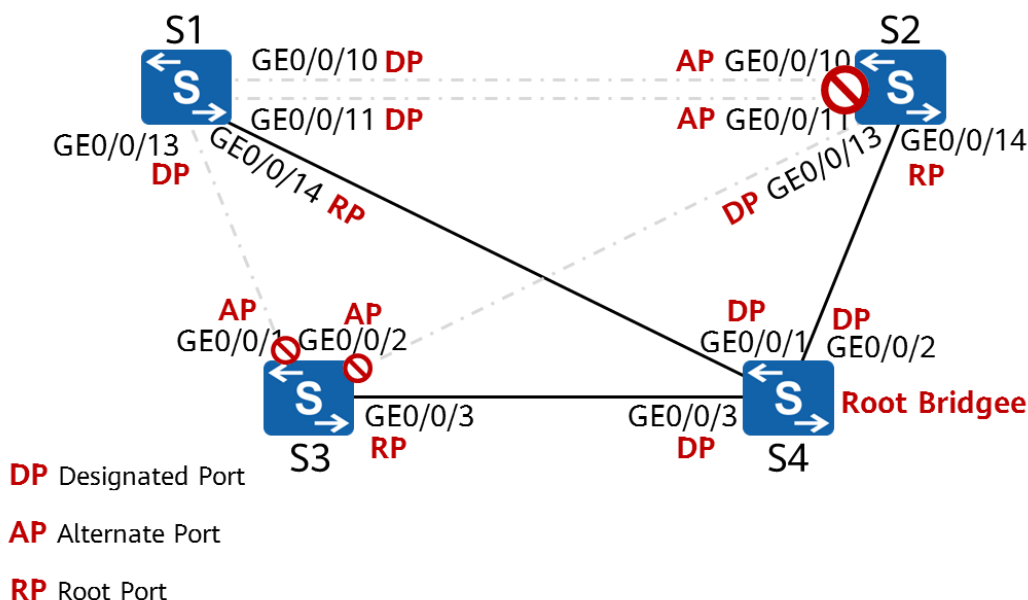
```
[S3]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/1    ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/2    ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/3    ROOT    FORWARDING   NONE
```

```
[S4]display stp brief
```



MSTID	Port	Role	STP State	Protection
0	GigabitEthernet0/0/1	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/2	DESI	FORWARDING	NONE
0	GigabitEthernet0/0/3	DESI	FORWARDING	NONE

- Based on the root bridge ID and port information on each switch, the current topology is as follows:



A dotted line indicates that a link does not forward service data. Note: This topology is for reference only and may not be the same as the actual spanning tree topology in the lab environment.

3.3 Modifying Device Parameters to Have S1 Become the Root Bridge and S2 the Secondary Root Bridge

- Change the bridge priorities of S1 and S2.

```
[S1]stp root primary
```

Owing to the importance of the root bridge, the switching device with high performance and network hierarchy is generally chosen as the root bridge. The priority of such a switching device, however, may not be that high. Therefore, you can run the corresponding command to configure the switching device as the root bridge to ensure that it becomes the root bridge. The **stp root** command configures a switching device as the root bridge or secondary root bridge of a specified spanning tree.

- If you run the **stp root primary** command to configure the current switching device as the root bridge in a specified spanning tree, the priority value of the device in the spanning tree is 0 and cannot be changed.
- If you run the **stp root secondary** command to configure the current switching device as the secondary root bridge in a specified spanning tree, the priority value of the switching device in the spanning tree is 4096 and cannot be changed.

```
[S2]stp root secondary
```



2. Check the STP status on S1.

```
[S1]display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge          :0      .4c1f-cc33-7359                //Bridge ID of the
device
Config Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times         :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC      :0      .4c1f-cc33-7359 / 0 //Current root bridge ID and root path
cost
CIST RegRoot/IRPC    :0      .4c1f-cc33-7359 / 0
CIST RootPortId      :0.0
BPDU-Protection      :Disabled
CIST Root Type       :Primary root
TC or TCN received   :84
TC count per hello   :0
STP Converge Mode    :Normal
Time since last TC    :0 days 0h:1m:44s
Number of TC         :21
Last TC occurred     :GigabitEthernet0/0/10
```

In this case, the bridge ID of S1 is the same as the root bridge ID, and the root path cost is 0, indicating that S1 is the root bridge of the current network.

3. Check the brief STP status information on all devices.

```
[S1]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/10  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/11  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/13  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/14  DESI    FORWARDING   NONE

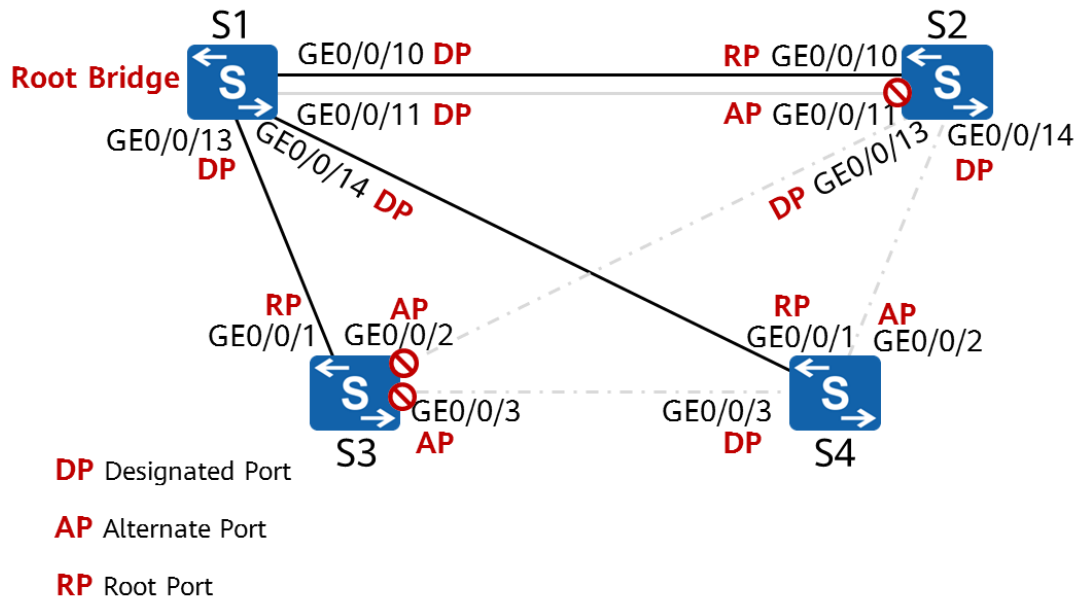
[S2]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/10  ROOT    FORWARDING   NONE
0      GigabitEthernet0/0/11  ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/13  DESI    FORWARDING   NONE
0      GigabitEthernet0/0/14  DESI    FORWARDING   NONE

[S3]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/1    ROOT    FORWARDING   NONE
0      GigabitEthernet0/0/2    ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/3    ALTE    DISCARDING   NONE

[S4]display stp brief
MSTID  Port                Role    STP State    Protection
0      GigabitEthernet0/0/1    ROOT    FORWARDING   NONE
0      GigabitEthernet0/0/2    ALTE    DISCARDING   NONE
0      GigabitEthernet0/0/3    DESI    FORWARDING   NONE
```



- Based on the root bridge ID and port information on each switch, the current topology is as follows:



3.4 Modifying Device Parameters to Have GigabitEthernet0/0/2 of S4 Become the Root Port

- Check the STP status on S4.

```
[S4]display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge           :32768.4c1f-cc10-5913
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC       :0 .4c1f-cc33-7359 / 20000
CIST RegRoot/IRPC     :32768.4c1f-cc10-5913 / 0
CIST RootPortId       :128.1
BPDU-Protection       :Disabled
TC or TCN received    :93
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:9m:5s
Number of TC          :18
Last TC occurred      :GigabitEthernet0/0/1
```

The cost of the root path from S4 to S1 is 20000.

- Change the STP cost of GigabitEthernet0/0/1 on S4 to 50000.

```
[S4]interface GigabitEthernet 0/0/1
[S4-GigabitEthernet0/0/1]stp cost 50000
```

- Check the brief STP status information.

```
[S4]display stp brief
```

MSTID	Port	Role	STP State	Protection
-------	------	------	-----------	------------



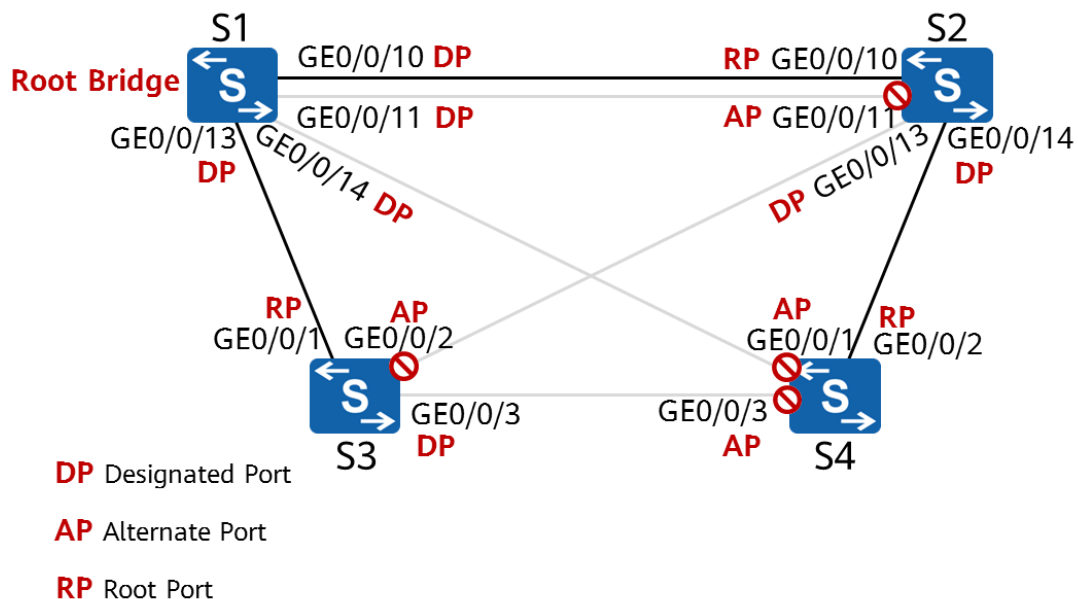
0	GigabitEthernet0/0/1	ALTE	DISCARDING	NONE
0	GigabitEthernet0/0/2	ROOT	FORWARDING	NONE
0	GigabitEthernet0/0/3	ALTE	DISCARDING	NONE

GigabitEthernet0/0/2 on S4 has become the root port.

4. Check the current STP status information.

```
[S4]display stp
-----[CIST Global Info][Mode STP]-----
CIST Bridge           :32768.4c1f-cc10-5913
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC      :0    .4c1f-cc33-7359 / 40000 //The root path cost is 40000 (=20000 + 20000).
CIST RegRoot/IRPC     :32768.4c1f-cc10-5913 / 0
CIST RootPortId       :128.2
BPDU-Protection        :Disabled
TC or TCN received    :146
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:2m:25s
Number of TC          :20
Last TC occurred      :GigabitEthernet0/0/2
```

5. The current topology is as follows:



3.5 Changing the Spanning Tree Mode to RSTP

1. Change the spanning tree mode on all devices.

```
[S1]stp mode rstp
Info: This operation may take a few seconds. Please wait for a moment...done.

[S2]stp mode rstp
Info: This operation may take a few seconds. Please wait for a moment...done.
```



```
[S3]stp mode rstp
Info: This operation may take a few seconds. Please wait for a moment...done.
```

```
[S4]stp mode rstp
Info: This operation may take a few seconds. Please wait for a moment...done.
```

2. Check the spanning tree status. S1 is used as an example.

```
[S1]display stp
-----[CIST Global Info][Mode RSTP]-----
CIST Bridge           :0       .4c1f-cc33-7359
Config Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
Active Times          :Hello 2s MaxAge 20s FwDly 15s MaxHop 20
CIST Root/ERPC        :0       .4c1f-cc33-7359 / 0
CIST RegRoot/IRPC     :0       .4c1f-cc33-7359 / 0
CIST RootPortId       :0.0
BPDU-Protection        :Disabled
CIST Root Type        :Primary root
TC or TCN received    :89
TC count per hello    :0
STP Converge Mode     :Normal
Time since last TC    :0 days 0h:0m:44s
Number of TC          :27
Last TC occurred      :GigabitEthernet0/0/11
```

After the spanning tree mode is changed, the topology of the spanning tree is not affected.

3.6 Configuring Edge Ports

GigabitEthernet0/0/10 to GigabitEthernet0/0/24 of S3 are connected only to terminals and need to be configured as edge ports.

```
[S3]interface range GigabitEthernet 0/0/10 to GigabitEthernet 0/0/24
```

A device typically provides multiple Ethernet ports, many of which have the same configuration. Configuring these Ethernet ports individually is tedious and error-prone. An easy way is to add such ports to a port group and configure the group. The system will then automatically execute the commands on all ports in the group.

Note: This function may not be available on some products.

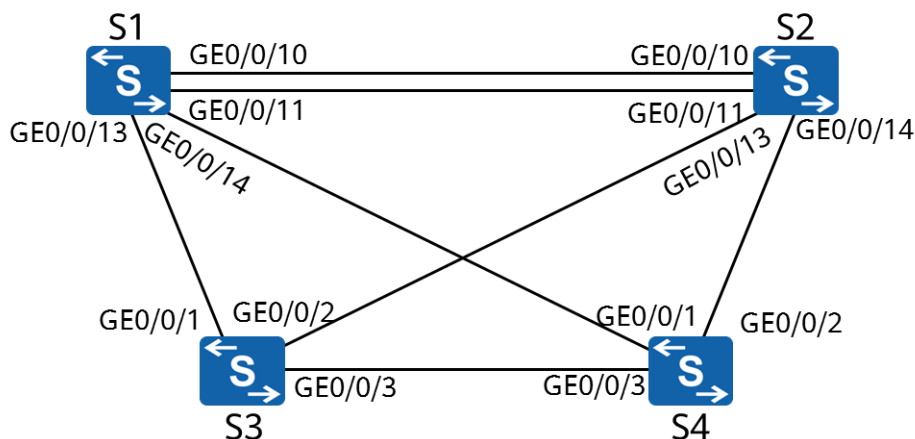
```
[S3-port-group]stp edged-port enable
```

The **stp edged-port enable** command sets the current port as an edge port. If an edge port of a switch receives a BPDU, the switch will automatically set the port as a non-edge port and recalculate the spanning tree.



4 Verification

1. Mark the root bridge and the role of each port in the lab environment based on the actual network convergence.



2. Disable any port on any switch and check whether the traffic can reach all other switches through the backup links.

5 Quiz

1. In Step 3, if the cost of GigabitEthernet 0/0/14 on S1 is changed to 50000, can the expected result be achieved? Why?
2. In the current topology, modify the configuration to have GigabitEthernet0/0/11 of S2 become the root port.
3. Can the two links between S1 and S2 be in the forwarding state at the same time? Why?